

PONKRISHNAN P

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PROFESSIONAL SUMMARY

AI & Data Science graduate with hands-on experience building LLM-powered pipelines, RAG systems, and production-grade REST APIs in Python. Familiar with LangChain, LangGraph, FastAPI, vector search, and prompt engineering fundamentals. Built real-world agentic and diagnostic AI systems from scratch. Comfortable working end-to-end from data ingestion to API exposure.

TECHNICAL SKILLS

- **Languages:** Python (proficient) | JavaScript | Java | SQL | C++ | TypeScript basics
- **AI & LLM:** LangChain | LangGraph | RAG Pipelines | Prompt Engineering | LLM Embeddings | Vector Search | NLP
- **ML Libraries:** PyTorch | Scikit-learn | NumPy | Pandas | OpenCV | TensorFlow
- **Frameworks & APIs:** FastAPI | REST/JSON | Spring Boot | React.js | Node.js | HTML | CSS
- **Data & Cloud:** PostgreSQL | MongoDB | MySQL | AWS (EC2, S3) | Docker | Git | CI/CD | Agile

WORK EXPERIENCE

DarwinBox — Product Development Intern, Hyderabad Jan 2026 – Apr 2026

- Built and exposed backend capabilities via FastAPI endpoints for enterprise HR and payroll modules, supporting ~10,000+ employee records across multi-tenant environments.
- Wrote evaluation and regression tests (PyTest) to monitor API and pipeline performance, improving code coverage by ~25% and reducing production defects.
- Optimized SQL queries and data workflows, cutting average retrieval time by ~30%; added structured logging and telemetry to track service health in production.
- Collaborated with senior engineers in Agile sprints — participated in design reviews, backlog refinement, and CI/CD deployments.

PROJECTS

AutoGenX – LLM Agentic Diagnostic Platform Python | FastAPI | LangChain | RAG | LLMs | MongoDB | React.js

- Built an LLM-powered agentic platform that processes real-time vehicle sensor data and routes diagnostic tasks using LangChain agent workflows.
- Implemented full RAG pipelines — document chunking, embedding generation, vector retrieval — to surface relevant diagnostic insights from unstructured operational data.
- Developed and exposed agent capabilities via FastAPI endpoints, enabling downstream tools and dashboards to consume structured diagnostic outputs via REST/JSON.
- Engineered prompt templates and LLM context management to improve answer accuracy and reduce hallucinations in diagnostic response generation.

DriveGuard – Real-Time AI Safety Monitoring System Python | OpenCV | PyTorch | Scikit-learn | REST APIs | MongoDB

- Designed an ML-powered anomaly detection system identifying driver fatigue and distraction in real time using computer vision and classification models built with PyTorch and Scikit-learn.
- Engineered multivariate feature extraction pipelines over behavioral time-series signals (eye closure rate, head pose, reaction time) feeding trained classification models.
- Built event-driven REST API services for low-latency alert delivery; wrote evaluation tests to validate detection accuracy and monitor model performance across test sets.

Enterprise HR Intelligence Platform – AI Data Layer Java | Spring Boot | PostgreSQL | AWS

- Designed ETL pipelines and optimized PostgreSQL queries improving throughput by ~40%; built secure REST APIs with RBAC and KPI engines for automated workforce analytics.

EDUCATION

B.Tech in Artificial Intelligence and Data Science 2022 – 2026

Sri Eshwar College of Engineering (Anna University), Coimbatore CGPA: 7.5+ | No Backlogs

Relevant Coursework: Machine Learning, Deep Learning, NLP, LLMs & Generative AI, Statistics, Computer Vision, DSA, DBMS

CERTIFICATIONS

- **AWS Cloud Quest: Cloud Practitioner** — Amazon Web Services [\[View\]](#)
- **NPTEL Cloud Computing (Elite)** — IIT Madras [\[View\]](#)
- **Java Programming & DSA** — Udemy [\[View\]](#)

ACHIEVEMENTS

- Solved **450+ DSA problems on LeetCode** across arrays, trees, graphs, and DP;
- selected for **DarwinBox's competitive internship program** contributing to production-grade enterprise software.